

Design of Continuous and Discrete LQI Control Systems with Stable Inner Loops

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Abstract: Design approaches were proposed for both continuous and discrete LQI (linear quadratic with integral) controllers, if exist, to stabilize the inner loops in addition to stabilizing the closed loops and minimizing the quadratic cost functionals. Derived from the algebraic Riccati equations involved in continuous and discrete LQI control, the design approaches were straightforwardly obtained by setting the quadratic cost functionals to decoupled forms with positive definite weighting matrices. Examples were provided to verify the effectiveness of the approaches.

Key words: linear quadratic with integral; algebraic Riccati equations; positive definiteness; Lyapunov stability

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Introduction

LQI (linear quadratic with integral) controllers are the static feedback controllers based upon plant state and integral (for continuous plants) or sum (for discrete plants) of tracking error. They stabilize the outer loops and minimize the quadratic cost functionals associated with step exogenous inputs, e.g., step reference input and step disturbance^[1,2]. LQI control systems have been applied to a wide area, e.g., motion and vibration control of three dimensional flexible shaking table^[3], electro-dynamic suspension control^[4,5], man-machine collaborated robotic systems^[6], dynamic positioning of ocean floating platform^[7] and shuttle tankers^[8], depth and heading control of unmanned underwater vehicles^[9], navigating of tunneling robots^[10], and attitude/position control of unmanned air vehicles^[11].

However, to the best of our knowledge, the problem of synthesizing LQI controllers which sta-

bilize the inner loops remains unresolved. The main reason to stabilize the inner loops is that their instability results in the closed loops of poor dynamic performance. As stated by Middleton^[12], for stable unit feedback systems, unstable open-loop poles impose fundamental constraints on achievable performance of closed loops. This is true with LQI control systems since the outer loops are based upon unit feedback and their stability is guaranteed by LQI design. Moreover, their open-loop stability is equivalent to that of the inner loops. Therefore, in order to remove the fundamental constraints on the performance of the outer loops caused by the open-loop instability, it is always desirable for LQI controllers to stabilize the inner loops.

The present paper presents the design approaches for synthesizing such LQI controllers for both continuous and discrete plants. The key idea of the design approaches is to utilize the freedom in choosing weighting matrices to achieve stable inner loops for the LTI (linear time-invariant) plants which can be stabilized by LQI controllers.

1 Continuous Case

Consider a continuous LTI plant with state-space realization

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$$\begin{cases} \dot{x}(t) = Ax(t) + Bu(t) \\ y(t) = Cx(t) + Du(t) \end{cases} \quad (1)$$

where $x(t) \in \mathbf{R}^n$, $u(t) \in \mathbf{R}^m$ and $y(t) \in \mathbf{R}^m$ are plant state, control and output vector, respectively, and A , B , C are matrices of proper dimensions

Define error-integral

$$\epsilon(t) = \int (r(\tau) - y(\tau)) d\tau$$

as the new state:

$$\dot{\tilde{\epsilon}}(t) = r(t) - Cx(t) \quad (2)$$

where $r(t) \in \mathbf{R}^m$ is reference input

Combining Eqs (1) and (2) yields the augmented state equation:

$$\begin{pmatrix} \dot{\tilde{x}}(t) \\ \dot{\tilde{\epsilon}}(t) \end{pmatrix} = \begin{pmatrix} A & 0 \\ -C & 0 \end{pmatrix} \begin{pmatrix} \tilde{x}(t) \\ \tilde{\epsilon}(t) \end{pmatrix} + \begin{pmatrix} B \\ 0 \end{pmatrix} u(t) + \begin{pmatrix} 0 \\ I_m \end{pmatrix} r(t) \quad (3)$$

where I_m is the unit matrix

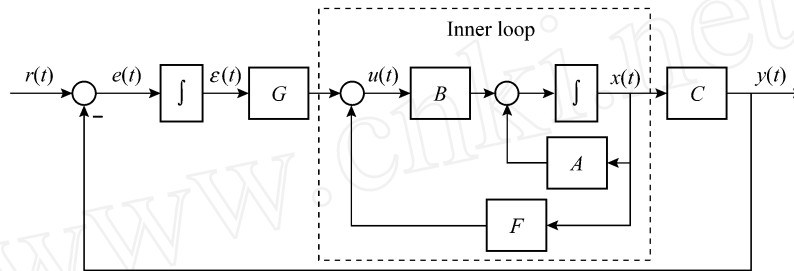


Fig 1 Continuous LQ I control system

Since $\dot{x}(\infty) = Ax(\infty) + Bu(\infty)$ and $\dot{\tilde{\epsilon}}(\infty) = r - Cx(\infty) = 0$ under step reference input, the augmented plant Eq (3) is equivalent to

$$\begin{pmatrix} \dot{\tilde{x}}(t) \\ \dot{\tilde{\epsilon}}(t) \end{pmatrix} = \begin{pmatrix} A & 0 \\ -C & 0 \end{pmatrix} \begin{pmatrix} \tilde{x}(t) \\ \tilde{\epsilon}(t) \end{pmatrix} + \begin{pmatrix} B \\ 0 \end{pmatrix} \tilde{u}(t) \quad (6)$$

and thus the continuous LQ I controller is actually a continuous LQ controller which stabilizes the plant Eq (6) and minimizes the cost functional Eq (5). Therefore with the notations

$$\hat{A} = \begin{pmatrix} A & 0 \\ -C & 0 \end{pmatrix}, \quad \hat{B} = \begin{pmatrix} B \\ 0 \end{pmatrix} \quad (7)$$

it is well known that^[13]: ① the continuous LQ I controller exists if (A, B) is stabilizable, $R = R^T > 0$, $Q = Q^T \geq 0$ and (Q, A) have no unobservable modes on the imaginary axis; ② the optimal feedback gain matrix is given by

$$(F \ G) = -R^{-1} \hat{B}^T \hat{P} \quad (8)$$

where $P = P^T \geq 0$ is the solution to the algebraic Riccati equation:

The continuous LQ I controller (see Fig 1), if exists, is an optimal state-feedback controller

$$u(t) = Fx(t) + G\epsilon(t) \quad (4)$$

which stabilizes the augmented plant Eq (3) and minimize the quadratic cost functional

$$J_c = \int_0^\infty [(\tilde{x}(t)^T \tilde{\epsilon}(t)^T) Q (\tilde{x}(t)^T \tilde{\epsilon}(t)^T)^T + \tilde{u}(t)^T R \tilde{u}(t)] dt \quad (5)$$

for step reference input, i.e. $r(t) \equiv r(t > 0)$, where $F \in \mathbf{R}^{n \times n}$ and $G \in \mathbf{R}^{m \times m}$ are feedback gain matrices of plant state and error-integral, respectively. $Q \in \mathbf{R}^{(n+m) \times (n+m)}$ and $R \in \mathbf{R}^{m \times m}$ are weighting matrices,

$$\tilde{x}(t) = x(t) - x(\infty), \quad \tilde{\epsilon}(t) = \epsilon(t) - \epsilon(\infty)$$

$$\tilde{u}(t) = u(t) - u(\infty)$$

$x(\infty)$, $\epsilon(\infty)$, $u(\infty)$ are steady-state values of plant state, error-integral and control, respectively.

$$P\hat{A} + \hat{A}^T P - P\hat{B}R^{-1}\hat{B}^T P + Q = 0 \quad (9)$$

Although the LQ I controller Eq (4), if exists, stabilizes the plant Eq (3), i.e. $\begin{pmatrix} A + BF & BG \\ -C & 0 \end{pmatrix}$ is stable, $(A + BF)$ may be unstable. In other words, the inner loop may be unstable (see Fig 1).

To analyze stability of the inner loop, the matrices P and Q are partitioned to comply with Eq (7), i.e.

$$P = \begin{pmatrix} P_{11} & P_{12} \\ P_{12}^T & P_{22} \end{pmatrix}, \quad Q = \begin{pmatrix} Q_{11} & Q_{12} \\ Q_{12}^T & Q_{22} \end{pmatrix} \quad (10)$$

where P_{11} , $Q_{11} \in \mathbf{R}^{n \times n}$, P_{12} , $Q_{12} \in \mathbf{R}^{n \times n}$ and P_{22} , $Q_{22} \in \mathbf{R}^{m \times m}$. In terms of Eqs (7), (8) and (10), the optimal feedback gain matrix of the inner loop is

$$F = -R^{-1} \hat{B}^T \hat{P}_{11} \quad (11)$$

Since there is freedom in choosing the weighting matrices Q and R in LQ I design for a given plant, a natural idea is to utilize the freedom to

achieve stable inner loops if possible and this leads to a sufficient condition for the LQ I controller to stabilize the inner loop.

Theorem 1 For a continuous LTI plant, suppose that (A, B) is stabilizable, $R = R^T > 0$ and $Q = Q^T > 0$, and $Q = \text{diag}(Q_{11}, Q_{22})$ with $Q_{11} \in \mathbf{R}^{n \times n}$ and $Q_{22} \in \mathbf{R}^{m \times m}$, then the LQ I controller exists and stabilizes the inner loop.

Proof Since $Q = Q^T > 0$, one has

$$\text{rank}(\lambda I_{n+m} - A - Q) = \text{rank}(Q) = n + m$$

where λ denotes any eigenvalue of A , (Q, A) is observable. Also take into account the stabilizability of (A, B) and the symmetry and positive definiteness of R , the LQ I controller exists since the existence conditions have been fully satisfied.

Moreover, the observability of (Q, A) guarantees the positive definiteness of the solution to the algebraic Eq (9)^[13], i.e. $P = P^T > 0$. In terms of Eq (10) and Schur complement lemma, one has $P_{22} > 0$ and $\bar{P}_{11} = P_{11} - P_{12}P_{22}^{-1}P_{12}^T > 0$.

Substitute Eqs (7), (10) and $Q = \text{diag}(Q_{11}, Q_{22})$ into Eq (9), it can be obtained that

$$\bar{P}_{11}A + A^T \bar{P}_{11} - \bar{P}_{11}B R^{-1} B^T \bar{P}_{11} + Q_{11} + P_{12}P_{22}^{-1}Q_{22}P_{22}^{-1}P_{12}^T = 0 \quad (12)$$

$$Q_{22} = P_{12}B R^{-1} B^T P_{12}^T \quad (13)$$

where the positive definiteness of P_{22} and $\bar{P}_{11}$ has been utilized. Inserting Eq (13) into Eq (12) yields

$$\bar{P}_{11}(A - B R^{-1} B^T P_{12}^T \bar{P}_{11}) + (A - B R^{-1} B^T P_{12}^T \bar{P}_{11})^T \bar{P}_{11} + P_{12}B R^{-1} B^T P_{12}^T \bar{P}_{11} + Q_{11} = 0 \quad (14)$$

i.e.

$$\bar{P}_{11}(A + B F) + (A + B F)^T \bar{P}_{11} + P_{12}B R^{-1} B^T P_{12}^T \bar{P}_{11} + Q_{11} = 0 \quad (15)$$

Due to positive definiteness of Q_{11} , R and P_{11} , one has

$$\bar{P}_{11}(A + B F) + (A + B F)^T \bar{P}_{11} < 0 \quad (16)$$

Since $\bar{P}_{11}$ is positively definite, $(A + B F)$ is stable according to Lyapunov's stability theorem, therefore the inner loop is stable and this concludes the proof.

Since stabilizability of $(\hat{A}, \hat{B})$ implies stabilizability of (A, B) and non-singularity of $\begin{bmatrix} A & B \\ C & 0 \end{bmatrix}$ ^[2], Theorem 1 suggests a design approach

to synthesize continuous LQ I controllers which stabilize the inner loops for a given LTI plant with (A, B) stabilizable and $\begin{bmatrix} A & B \\ C & 0 \end{bmatrix}$ non-singular: set the quadratic cost functional in a decoupled form with positively definite weighting matrices, i.e.

$$J_c = \int [\hat{x}(t)^T Q_{11} \hat{x}(t) + \hat{e}(t)^T Q_{22} \hat{e}(t) + \hat{u}(t)^T R \hat{u}(t)] dt \quad (17)$$

where $Q_{11} = Q_{11}^T > 0$, $Q_{22} = Q_{22}^T > 0$ and $R = R^T > 0$.

The following example illustrates the effectiveness of the design approach.

Example 1 Design an LQ I controller which stabilizes the inner loop for an LTI plant with state-space matrices

$$A = \begin{bmatrix} 0 & 541 & 3 & 0 & 043 & 4 & 0 & 756 & 8 \\ 0 & 693 & 8 & 0 & 090 & 1 & 0 & 631 & 5 \\ 0 & 078 & 7 & 0 & 142 & 6 & 0 & 617 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$C = (1 \quad 0 \quad 0)$$

It can be easily checked that (A, B) is stabilizable and $\begin{bmatrix} A & B \\ C & 0 \end{bmatrix}$ is non-singular. Therefore the

LQ I controller exists for appropriate weighting matrices. While the control weighting is simply chosen as $R = 1$, two different choices of the weighing matrix Q are presented for comparison, i.e.

$$Q^1 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}, \quad Q^2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Note that whereas the sufficient conditions for the LQ I controller to exist and stabilize the inner loop are all met with $Q = Q^2$, only existence conditions of LQ I controllers are satisfied with $Q = Q^1$. The resulting LQ I gain matrices corresponding to $Q = Q^1$ and $Q = Q^2$ are

$$F^1 = (-3.5495 \quad -4.7734 \quad -19.9028)$$

$$F^2 = (-3.9669 \quad -5.7594 \quad -28.2727)$$

$$G^1 = -1, \quad G^2 = -1$$

respectively, and the poles of the associated inner loops are

$$\lambda(A + B F^1) = (0.0495, -1.1752 \pm 1.1668i)$$

$$\lambda(A + B F^2) = (-0.0703, -1.3240 \pm 1.3170i)$$

Obviously the selection $Q = Q^2$ yields a stable inner loop while the selection $Q = Q^1$ does not.

The step responses of both LQ I control systems are shown in Fig 2. It can clearly be seen that the dynamic performance of the LQ I control system with stable inner loop is superior to that with unstable inner loop.

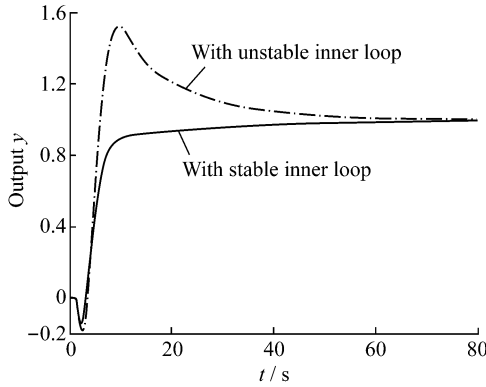


Fig 2 Step response of continuous LQ I control system

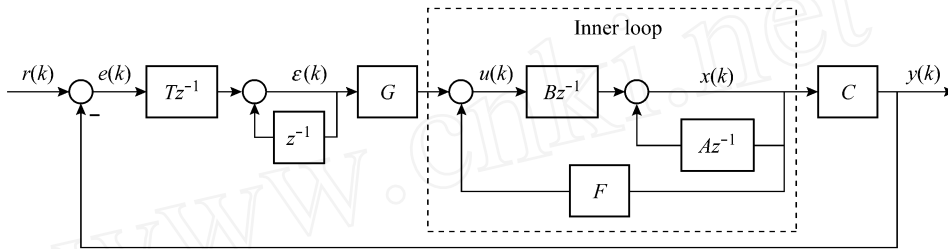


Fig 3 Discrete LQ I control system

The discrete LQ I controller (see Fig 3), if exists, is a state-feedback controller:

$$u(k) = Fx(k) + G\epsilon(k) \quad (21)$$

which stabilizes the augmented plant Eq (20) and minimizes the quadratic cost functional:

$$J_d = \sum_{k=0}^{\infty} [(\tilde{x}(k)' \quad \tilde{\epsilon}(k)') Q (\tilde{x}(k)' \quad \tilde{\epsilon}(k)') + \tilde{u}(k)' R \tilde{u}(k)] \quad (22)$$

for step reference input, i.e. $r(k) \equiv r(k \geq 0)$, where Q, R are weighting matrices, $\tilde{x}(k) = x(k) - x(\infty)$, $\tilde{\epsilon}(k) = \epsilon(k) - \epsilon(\infty)$, $\tilde{u}(k) = u(k) - u(\infty)$, and $x(\infty), \epsilon(\infty), u(\infty)$ are steady-state values of plant state, error-integral and control, respectively.

Since $x(\infty) = Ax(\infty) + Bu(\infty)$ and $\epsilon(\infty) = \epsilon(\infty) + r - Cx(\infty)$ under step reference input, the augmented plant Eq (20) is equivalent to

$$\begin{bmatrix} \tilde{x}(k+1) \\ \tilde{\epsilon}(k+1) \end{bmatrix} = \begin{bmatrix} A & 0 \\ -C & I_m \end{bmatrix} \begin{bmatrix} \tilde{x}(k) \\ \tilde{\epsilon}(k) \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} \tilde{u}(k) \quad (23)$$

and the discrete LQ I controller is a discrete LQ

2 Discrete Case

Consider discrete LTI plant with state-space realization

$$\begin{cases} x(k+1) = Ax(k) + Bu(k) \\ y(k) = Cx(k) \end{cases} \quad (18)$$

Define error-sum as

$$\epsilon(k+1) = \epsilon(k) + T[r(k) - Cx(k)] \quad (19)$$

where T is the sampling period and is assumed to be 1 s hereafter without loss of generality.

Combining Eqs (18) and (19) generates the augmented plant

$$\begin{bmatrix} x(k+1) \\ \epsilon(k+1) \end{bmatrix} = \begin{bmatrix} A & 0 \\ -C & I_m \end{bmatrix} \begin{bmatrix} x(k) \\ \epsilon(k) \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} u(k) + \begin{bmatrix} 0 \\ I_m \end{bmatrix} r(k) \quad (20)$$

controller which stabilizes the plant Eq (23) and minimizes the cost functional Eq (22). Hence with the notations

$$\bar{A} = \begin{bmatrix} A & 0 \\ -C & I_m \end{bmatrix}, \quad \bar{B} = \begin{bmatrix} B \\ 0 \end{bmatrix} \quad (24)$$

it is well known that: ① the discrete continuous LQ I controller exists if $(\bar{A}, \bar{B})$ is stabilizable, $R = R^T > 0$, $Q = Q^T \geq 0$ and $(Q, \bar{A})$ have no unobservable modes on the unit circle; ② the optimal feedback gain matrix is given by

$$(F \ G) = -(\bar{B}^T P \bar{B} + R)^{-1} \bar{B}^T P \bar{A} \quad (25)$$

where $P = P^T \geq 0$ is the solution to the algebraic Riccati equation:

$$\bar{A}^T P \bar{A} - P - \bar{A}^T P \bar{B} (\bar{B}^T P \bar{B} + R)^{-1} \bar{B}^T P \bar{A} = 0 \quad (26)$$

Although the LQ I controller Eq (21) stabilizes the augmented plant Eq (20), i.e. $\begin{bmatrix} A + BF & BG \\ -C & I_m \end{bmatrix}$ is stable, $(A + BF)$ may be unstable. In other words, a discrete LQ I controller may not stabilize the inner loop (cf Fig 3).

To analyze stability of the inner loop, as in Eq (10) the matrices P and Q are partitioned to comply with Eq (24). In terms of Eqs (24), (25) and (10), the feedback gain matrix of the inner loop can be written as

$$F = -(B' P_{11} B + R)^{-1} B' (P_{11} A - P_{12} C) \quad (27)$$

Similar to continuous case, the freedom in choosing Q can be utilized to synthesize the discrete LQ I controllers which stabilize the inner loops

Theorem 2 For a discrete LTI plant, suppose that $(\bar{A}, \bar{B})$ is stabilizable, $R = R' > 0$ and $Q = \text{diag}(Q_{11}, Q_{22})$ is symmetric and positively definite, then the discrete LQ I controller exists and stabilizes the inner loop.

Proof Since $Q = Q' > 0$, one has $\text{rank}(\lambda I_{n+m} - \bar{A}' Q) = \text{rank}(Q) = n+m$ (λ is any eigenvalue of $\bar{A}$), therefore $(Q, \bar{A}')$ is observable. Take into account the stabilizability of $(\bar{A}, \bar{B})$ and symmetry and positive definiteness of R , the existence conditions of discrete LQ I controller are all met. Therefore the discrete LQ I controller exists

Furthermore, the observability of $(Q, \bar{A}')$ implies the positive definiteness of P . From its partition Eq (10) and Schur complementary lemma, one has $P_{22} > 0$ and $\bar{P}_{11} = P_{11} - P_{12} P_{22}^{-1} P_{12}' > 0$.

According to Eq (25),

$$\bar{A}' P B (B' P B + R)^{-1} B' P A = - (F' G) (B' P B + R) (F' G) - \bar{A}' P B (F' G) - (F' G) B' P A \quad (28)$$

By inserting Eqs (24), (10), (28) and $Q = \text{diag}(Q_{11}, Q_{22})$ into Eq (26), the (1, 1) and (2, 2) blocks of the algebraic Riccati equation can be written as

$$(A + BF)' P_{11} (A + BF) + C' P_{22} C - (A + BF)' P_{12} C - C' P_{12}' (A + BF) - P_{11} + F' R F + Q_{11} = 0 \quad (29)$$

$$(A + BF)' (P_{11} B G + P_{12}) - C' (P_{12}' B G + P_{22}) - P_{12} + F' R G = 0 \quad (30)$$

In terms of Eqs (27) and (30) one can obtain

$$C = P_{22}^{-1} P_{12}' [(A + BF) - I_n] \quad (31)$$

Replacing C in Eq (29) with Eq (31) yields

$$(A + BF)' \bar{P}_{11} (A + BF) - \bar{P}_{11} + Q_{11} + F' R F = 0 \quad (32)$$

Due to positive definiteness of Q_{11} and, it is clear that

$$(A + BF)' \bar{P}_{11} (A + BF) - \bar{P}_{11} < 0 \quad (33)$$

Since $\bar{P}_{11}$ is also positively definite, $(A + BF)$ is stable according to Lyapunov stability theorem and this concludes the proof

Similar to continuous case, Theorem 2 suggests a design approach for synthesis of discrete LQ I controllers for a discrete plant with (A, B) stabilizable and $\begin{bmatrix} A - I_n & B \\ C & 0 \end{bmatrix}$ non-singular: set the quadratic cost functional in a decoupled form with positive definite weighting matrices, i.e.

$$J_d = \sum_{k=0}^{\infty} [\tilde{x}(k)' Q_{11} \tilde{x}(k) + \tilde{e}(k)' Q_{22} \tilde{e}(k) + \tilde{u}(k)' R \tilde{u}(k)] \quad (34)$$

The following example verifies the design approach

Example 2 Design a discrete LQ I controller which stabilizes the inner loop for a discrete LTI plant with state-space matrices:

$$A = \begin{bmatrix} 1.3798 & 0.3100 & -0.6509 \\ -0.0742 & -0.9057 & 0.8052 \\ 0.3207 & 0.7356 & 0.5949 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad C = (1 \quad 0 \quad 0)$$

It can be readily checked that (A, B) is stabilizable and $\begin{bmatrix} A - I_3 & B \\ C & 0 \end{bmatrix}$ is non-singular. Therefore the discrete LQ I controller exists for appropriate weighting matrices. The weighting matrices are the same as that in Example 1.

As in continuous case, while only existence conditions of discrete LQ I controllers are satisfied with $Q = Q^1$, the sufficient conditions for the discrete LQ I controller to exist and stabilize the inner loop are all met with $Q = Q^2$. The resulting discrete LQ I gain matrices corresponding to $Q = Q^1$ and $Q = Q^2$ are

$$F^1 = (-1.0695 \quad 2.0965 \quad -0.1108)$$

$$F^2 = (-1.1608 \quad 1.9757 \quad -0.3998)$$

$$G^1 = 0.3332, \quad G^2 = 0.3160$$

respectively, and the poles of the associated inner loops are

$$\lambda(A + BF^1) = (1.0006, -0.2689, -0.7321)$$

$$\lambda(A + BF^2) = (0.9081, -0.3141, -0.6858)$$

Obviously our design approach yields a stable inner loop while the conventional LQ I design approach

does not produce a stable inner loop.

The step responses of both LQ I control systems are shown in Fig 4. It can clearly be seen that the dynamic performance of the LQ I control system with stable inner loop is much better.

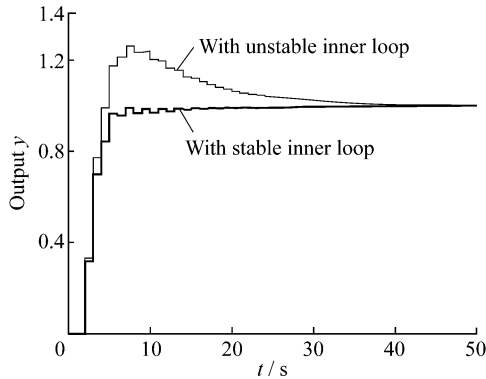


Fig 4 Step response of discrete LQ I control system

3 Conclusion

For both continuous and discrete LTI plants which can be stabilized by LQ I controllers, the stability of the inner loops involved in the LQ I control systems can be achieved by restricting the quadratic cost functionals to decoupled forms with positively definite weighting matrices. The future work will focus on experimental verifications of the proposed approach in the design of servo systems for ocean floating platforms.

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